LEARN MORE We've only given a brief sketch of the TLS handshake process and certificate authentication. An excellent resource for learning more is *Bulletproof SSL and TLS* by Ivan Ristić (Feisty Duck, 2015).

POP QUIZ

Answers at the end of the chapter.

5. To request client certificate authentication, the server must send which one of the following messages?

a) Certificate

b) ClientHello

c) ServerHello

d) CertificateVerify

e) CertificateRequest

6. How does TLS prevent a captured CertificateVerify message being reused for a different TLS session? Pick one answer.

a) The client's word is their honor.

b) The CertificateVerify message has a short expiry time.

c) The CertificateVerify contains a signature over all previous messages in the handshake.

d) The server and client remember all CertificateVerify messages they've ever seen.